

chelseas proejct: <https://devpost.com/software/test-5n3w8y>

Plan:

- US only
- Data sourceS:
 - flight data
 - international data - scrape news headlines for that
 - NOTAMs and TAFs - online regional data
 - weather data
- map with flights on it
- price estimate - see cost breakdown
- cost optimization
- see different options and simulate price for all (ex: how much fuel is used, how much time it takes)
- constrain to certain time frame and certain events

cost saved icon

- Three hours is a brutal crunch, but your concept is killer. To pull this off for a hackathon demo, your highest priority is building a **high-fidelity vertical slice**—meaning you shouldn't waste 2 hours wrestling with live API keys and edge-case data streams. Instead, hardcode an elite, true-to-life case study to show the judges exactly how the system works under pressure. The SpaceX event from last week is the absolute perfect anchor for your project.

Time and distance modeling

The Perfect Case Study: SpaceX Starship Flight 12 (May 22, 2026)

On **Friday, May 22, 2026**, SpaceX launched its massive Starship Version 3 rocket from Starbase in Boca Chica, Texas. While the launch itself was in Texas, the FAA issued a massive Temporary Flight Restriction (TFR) that extended deep into the Caribbean, completely shutting down parts of the **Kingston Flight Information Region (FIR)** between **5:30 PM and 7:43 PM EDT.**

What Happened in the Real World:

- **The Chaos:** JetBlue flight B61575 (Fort Lauderdale to Kingston) flew directly into the restricted window, was forced to pull a complete U-turn mid-flight, and diverted back to Miami.
- **The Casualties:** Multiple other international flights (Delta, British Airways, Cayman Airways) suffered massive ground delays or were forced to leave aircraft and crews overnight in Jamaica.
- **The METAR/TAF Context:** Weather stations at Miami (KMIA), Fort Lauderdale (KFL), and Kingston (MKJP) were reporting perfectly clear, beautiful VFR (Visual Flight Rules) conditions.

The Platform Wow-Factor: A legacy airline dashboard looking *only* at weather data would show green skies. Your platform, however, cross-references the space NOTAM polygon against scheduled routes, instantly flagging that JetBlue B61575 is flying into a logistical wall despite the perfect weather.

3-Hour Hackathon Action Plan

To win a hackathon, you need a working UI and a clear financial hook within minutes of starting your pitch. Here is how you should budget your next 180 minutes:

1.Hour 1: Hardcode the SpaceX 'Smoke Test' Data State:Build the Data Foundation.

Do not build a live web scraper right now. Mock a JSON data file representing the evening of May 22, 2026. Include standard flight tracks for 3-4 flights (like JetBlue B61575), mock "Clear/Sunny" weather JSON for KFL and MKJP, and a hardcoded coordinates array representing the active space TFR airspace restriction box.

2.Hour 2: Write the Simulation Logic & Cost Math:The Math Engine.

Build a simple JavaScript engine that calculates the financial impact of different operator decisions. Write basic functions that weigh fuel burn rates against hourly gate-delay fees and overnight crew accommodation costs.

3.Hour 3: Wire Up the Frontend & 'Cost Saved' Banner:The UI & Pitch Polish.

Render your mock flights on a lightweight map (like Leaflet.js or Mapbox) with a big red polygon showing the closed airspace. Create an interactive sidebar for the operator. When the judge clicks "Simulate Ground Hold," update the UI dynamically to flash a green "Cost Saved" badge.

Data Architecture & Pricing Factors Matrix

When you explain your platform architecture to technical judges, show them this exact breakdown of how disparate data sources map directly to airline operational costs:

Data Domain	Best API Source (Post-Hackathon)	Core Pricing Factor	Formula Variable to Simulate
Airspace Boundaries	FAA NOTAM Search / OpenSky Network	TFR En-Route Restrictions: Forces sudden horizontal or vertical bypass routes.	Extra miles flown $\times$ Jet-A fuel burn rate (~\$3.50+ per gallon).
Weather Status	NOAA / Aviation Weather Center (METAR/TAF)	Ground Stops / Low Ceiling: Restricts acceptance rates at arrival airports.	Holding pattern fuel burn (~\$2,000 to \$5,000 per hour depending on aircraft size).

Geopolitical Events	GDELT Project / NewsAPI	Regional Risk Escalations: Volatile local fuel pricing or sudden airport security closures.	Regional fuel surcharges and dynamic insurance premium spikes.
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The Operator Simulation Playbook

Your frontend sidebar should give the user three explicit "Actions" to resolve the SpaceX Caribbean airspace crisis. Your simulator will calculate the price for each:

Option 1: Proactive Ground Hold (The "Smart" Move)

- **What it does:** The operator catches the NOTAM conflict *before* takeoff and holds JetBlue B61575 on the tarmac at Fort Lauderdale for 2 hours until the space debris window clears.
- **Cost Variables:** Minimal fuel burn (APU auxiliary power unit only), low gate-use fees, minor passenger delay compensation.
- **Hackathon Impact:** This is your primary recommendation. It saves massive amounts of money compared to what actually happened.

Option 2: Active Reroute (The "Flexible" Move)

- **What it does:** The operator flies the plane on time but modifies the flight path to route completely around the eastern edge of the Kingston FIR restriction zone.
- **Cost Variables:** High extra fuel burn (adds ~45 minutes of flight time), additional flight crew hourly pay blocks.

Option 3: Blind Continuance & Forced Turn-Back (The Real-World Failure)

- **What it does:** The operator ignores the regional constraint. The plane flies deep into the Caribbean, finds the airspace locked down, pulls a mid-air turn-back, and diverts to Miami.
- **Cost Variables: Absolute worst case.** Double the fuel burn, emergency airport landing fees at Miami, hotel vouchers for 150+ stranded passengers, and expired crew duty-hour limits requiring a backup crew to be flown in.

The Pitch Punchline

When presenting, you can look at the judges and say: *"On May 22, airlines suffered massive losses because they didn't connect space NOTAMs with flight routing. By clicking our 'Proactive Ground Hold' button, our simulator proves that an operator could have spent \$4,000 in gate delays to save over \$45,000 in wasted fuel and emergency diversions. That is the green 'Cost Saved' icon you see blinking at the top right."*

Ideas

- 1
 - Sources
 - Applicability
- 2
 - Sources
 - Applicability
- 3

- Sources
 - Applicability
- 4
 - Sources
 - Applicability

Hello

command and control
 visualization and seeing the options, what happens if i do this

use operators to make decisions giving tool, using simulation data
 are planes in a bad area
 full cost on system if something happens - monetary cost
 rocket launches - close this airspace
 military artillery - if an airspace closes
 monitoring geopolitical events
 integrate live price of oil - price of a flight

3 stages of ops: strategic, tactical day of next 24 hrs how to modify plans and react, operational
 - in the moment
 pricing is strategic or tactical, airspace deconfliction is - operational
 pricing - effects of pricing

ability to look at state of world, monitor geopolitical events and TAFs, and making changes - if
 this happens what is the cost on the whole system, airspace closures - how to model that? gas
 prices? having to reroute?
 handling what if scenarios, make it a live application.

ATC - they have a world model, how much would it cost
 lay out the decisions - close airport or runway, we can tell you how much that would cost, give
 them an optimizer

exposure to gas prices - if iran closed
 he thinks thinking about pricing is super specific, closures. tying it into defense

Sources:

Tafts

